

-sketch:

NDA's, complex numbers, mention rotations, can we find 3d? quaternions, geometry, cayley-dickson, octonions, cross product, hurwitz's theorem, qed

Intro

An algebra is a vector space with a product:

Examples: $\mathbb{R}, \mathbb{C}$, real-valued functions (pointwise) product, hadamard product (useful in statistics, optimization, and random matrix theory). There are interesting examples that aren't commutative or associative though, for instance the cross product:

$$(a \times a) \times b = 0 \neq a \times (a \times b)$$

A division algebra is a (not-necessarily associative) algebra where we can do division, i.e. for nonzero $a \in A$, there exists $x \in A$ such that $ax = 1 = xa$. We can also look to equip our space with a norm and inner product, usually asking for compatibility with the inner product/norm: $\|ab\| = \|a\|\|b\|$. A good example to keep in mind is the complex numbers!

Cayley-Dickson

In particular, we can construct interesting normed division algebras in the following way: As a vector space structure, consider $\mathbb{R} \oplus \mathbb{R}$ (ordered pairs of real numbers), and give this the multiplication $(a, b)(c, d) = (ac - d^*b, da + bc^*)$. This is just $\mathbb{C}$. Define $(a, b)^* = (a^*, -b)$, where for reals, $a^* = a$, and we have division $(a, b)^{-1} = \frac{(a, b)^*}{((a, b)^*(a, b), 0)}$

Important properties:

- (1) Identify $\mathbb{R} = \text{Span}\{(1, 0)\}$ (so $(1, 0) = 1$).
- (2) $|(a, b)|^2 = (a, b)^*(a, b) = (a, b)(a, b)^*$, where $|\cdot|$ is the norm on $\mathbb{R}^2$, and one can check this is induced by the inner product by polarization, $\langle A, B \rangle = \frac{1}{4}(|A+B|^2 - |A|^2 - |B|^2)$ for $A, B \in \mathbb{C}$.
- (3) $(a, b)^{**} = (a, b)$
- (4) $|(a, b)(c, d)| = |(a, b)||c, d|$, from which it follows that $\langle ab, ab \rangle = \langle a, a \rangle \langle b, b \rangle$ by polarization.
- (5) To get an explicit construction for $\mathbb{R}^\perp$, it is 1-dimensional (since $\mathbb{C}$ is 2-dimensional and $\mathbb{R}$ is 1-dimensional). Pick a unit vector $i = (a, b) \in \mathbb{R}^\perp$, then $\mathbb{C} = \mathbb{R} \oplus \mathbb{R}i$. Then $(1, 0) = |i|^2 = (a, b)(a, -b) = (a^2 + b^2, -ab + ba) = (a^2 + b^2, 0)$, so $a^2 + b^2 = 1$, but $a = \langle (1, 0), i \rangle = 0$, so $b^2 = 1$ so $b = \pm 1$, so $i = (0, \pm 1)$, and we see $i^2 = (-1, 0) = -1$.
- (6) Suppose $x, y \in \text{Im } \mathbb{C} := \mathbb{R}^\perp$. Then $xy = -x \cdot y + 0$, this will be important later.

Notice to do this construction, we got a nice normed division algebra, but lost a nice algebraic property of $\mathbb{R}$, namely the natural order $<$!

Now, let's do the construction again. As a vector space construction, consider $\mathbb{C} \oplus \mathbb{C}$, we'll call this $\mathbb{H}$, and define the multiplication

$(a, b)(c, d) = (ac - d^*b, da + bc^*)$, where $a, b, c, d \in \mathbb{C}$. Define the involution $(a, b)^*$ as before, and one checks this induces a norm that agrees with the inner product in the same way, with $|(a, b)|^2 = (|a|^2 + |b|^2, 0)$ as before. As before, we can take $\mathbb{R} \cong \text{Span}\{(1, 0)\}$, and take $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}^\perp = \mathbb{R} \oplus \mathbb{R}i + \mathbb{R}j + \mathbb{R}k$ for i, j, k unit vectors orthogonal to $\mathbb{R}$ satisfying $i^2 = j^2 = k^2 = -1$. However, $\mathbb{H}$ isn't commutative (we lost another algebraic property!), we have $i = (0 + i, 0)$, $j = (0, 1)$, where

$$ij = (i, 0)(0, 1) = (0, 0 + i) = k$$

but

$$ji = (0, 1)(i, 0) = (0, 1i^*) = (0, 0 - i) = -k$$

(The rules $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$ with $i^2 = j^2 = k^2 = -1$ determine the quaternions, these are the right hand rule).

We'll return to the quaternions soon, but briefly, we can define the octonions $\mathbb{O}$ in the same way: Take $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}$, define multiplication and involution $(a, b)^*$ as before, this gives a norm, agreeing with inner product etc., $\mathbb{R} \cong \text{Span}\{1\}$, $\mathbb{O} = \mathbb{R} \oplus \mathbb{R}^\perp$ with $\mathbb{R}^\perp$ seven dimensional spanned by 7 square roots of -1 which are unit vectors, $e_1, \dots, e_7$.

This isn't even associative, $(e_1 e_2) e_3 = e_4 e_3 = -e_6$, $e_1 (e_2 e_3) = e_1 e_5 = e_6$, so we lost another algebraic property.

We'll stop here, (you can get the sedonions $\mathbb{O} \oplus \mathbb{O}$, but they aren't even alternating $(x(xy) \neq (xx)y)$, which in particular implies $|x||y| \neq |xy|$: (probably skip this proof). $|\cdot|$ is still induced by $*$ the way we defined it above, so $x + x^* = \langle x, 1 \rangle = \langle x, 1 \rangle 1$, and $x^* x y - x x^* y = |x|^2 y - |x|^2 y = 0$, so

$$0 = ((\langle x, 1 \rangle - x)x)y - x((\langle x, 1 \rangle - x)y) = (\langle x, 1 \rangle x - x^2)y - x(\langle x, 1 \rangle y - xy) = |x|^2 y - (xx)y - |x|^2 y - x(xy) = 0$$

so we lose the nice normed-ness of our division algebra.

So why do we care? Well as you know, complex numbers represent rotations in $\mathbb{R}^2$. William Hamilton (hence $\mathbb{H}$) was interested in a normed division algebra like $\mathbb{C}$ that talks about 3d rotations. The problem (as we'll sketch the proof of later) is that one doesn't exist! But we can still represent 3d rotations:

Let $u, v \in \text{Im}\mathbb{H} := \mathbb{R}^\perp \cong \mathbb{R}^3$. Then $uv = -u \cdot v + ((u_2 v_3 - v_3 u_2)i + (u_3 v_1 - u_1 v_3)j + (u_1 v_2 - v_2 u_1)) = -u \cdot v + (u \times v)$, identifying $\text{Im}\mathbb{H} \cong \mathbb{R}^3$ to get a cross product, so $u \times v = \text{Im}(uv)$ for imaginary u, v . More generally, $(a, u)(b, v) = (ab - u \cdot v, av + bu + u \times v)$.

But the cross product tells us about rotations in $\mathbb{R}^3$: any rotation in $\mathbb{R}^3$ is determined by an axis (unit vector) a , an angle α , and a vector b , the rotated vector c is

$$c = (1 - \cos \alpha)(a \cdot b)a + b \cos \alpha + (a \times b) \sin \alpha \quad (\text{Rodrigues' rotation formula})$$

So we see the cross product and dot product are enough to do rotations! But we know how the dot product and cross product show up in quaternion multiplication, so it should make intuitive sense that quaternion multiplication encodes this exact formula: Let q be the quaternion

$$q = \underbrace{\cos \alpha/2}_{\mathbb{R}} + \underbrace{a \sin \alpha/2}_{\text{Im}\mathbb{H} \cong \mathbb{R}^3}$$

(this is a tedious trig exercise using the formula for quaternion multiplication).

Then $c = qbq^{-1}$, where we note the importance of the noncommutativity (it makes sense, 3d rotations aren't commutative, for example! It should make sense these are 4d, a rotation actually has 4 dimensions to it: 3 dimensions to define an axis of rotation, 1 dimension to describe the angle rotation itself!

It turns out quaternions are used all the time in applications: the fact you only need 4 numbers to represent a rotation, and the fact that there's no roundoff error are super useful.

So we see quaternions elegantly describe 3-space rotations, essentially because they encode the cross product (and dot product).

Now, we saw previously that for $u, v \in \mathbb{R}^3 \cong \text{Im}\mathbb{H}$, $u \times v = \text{Im}(uv)$. What if we do the exact same thing to define a cross product in $\mathbb{R}^7$?

Well the cross product in $\mathbb{R}^3$ satisfies some nice properties, it's

- (1) bilinear
- (2) $u \times v$ is perpendicular (in the sense of the dot product) to u and v
- (3) We can predict the norm nicely,

$$|u \times v|^2 = |u|^2 |v|^2 \sin^2 \theta = |u|^2 |v|^2 (1 - \cos^2 \theta) = |u|^2 |v|^2 - |u|^2 |v|^2 \cos^2 \theta = |u|^2 |v|^2 - (u \cdot v)^2$$

It turns out if we define a cross product in $\mathbb{R}^7$ this way $u, v \in \mathbb{R}^7 \cong \text{Im}\mathbb{O}$, then $u \times v = \text{Im}(uv)$, it satisfies the same properties, so we meaningfully have a cross product in $\mathbb{R}^7$ (in particular, it gives a perpendicular vector from any pair!). This encodes rotations like the $\mathbb{R}^3$ one does, but the Rodrigues rotation formula is a lot more complicated.

These are the only cross products we can define, to see this:

Theorem (Hurwitz)

Let A be a finite dimension normed division algebra over $\mathbb{R}$ with norm induced by inner product. Then $A \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$.

IDEA:

Let B be a proper subalgebra of A containing 1. Pick $j \in B^\perp$ a unit vector, and define $C = \langle B, j \rangle = B \oplus Bj$. Then C satisfies the multiplication $(a+bj)^* = a^* - bj$, $(a+bj)(c+dj) = (ac - d^*b, da + bc^*)$. Then, using the restriction that A is a normed division algebra (COMPATIBLE WITH MULTIPLICATION!) on C , we get $\langle ac, d^*b \rangle = \langle bc^*, da \rangle$, but $\langle ac, d^*b \rangle = \langle d(ac), b \rangle$ and $\langle bc^*, da \rangle = \langle b, (da)c \rangle$, so since b can be arbitrary, this shows $(da)c = d(ac)$, so C must be associative.

Now, let A be a F.D. normed division algebra over $\mathbb{R}$. Then it contains $\text{Span}\{1\} \cong \mathbb{R}$, and if this is the whole thing, we are done. Otherwise, take $B = \mathbb{R}$, and this shows A contains $C \cong \mathbb{C}$ by Cayley-Dickson. If this is the whole thing, we are done, otherwise take $B = \mathbb{C}$ and then $C \cong \mathbb{H}$ by Cayley-Dickson. If this is the whole thing, we're done, otherwise take $B = \mathbb{H}$, and so $C = \mathbb{O}$. But here the process must stop, because we cannot take $B = \mathbb{O}$ because it isn't associative!!!! So these are all the possible cases. ■

Corollary

If $n \geq 3$, and $\times : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a cross product satisfying (1), (2), (3), then $n = 3$ or $n = 7$.

This can be weakened to the following three properties:

- (1) $a \times b$ is continuous
- (2) $a \times b$ is perpendicular to a and b
- (3) $a \times b \neq 0$ for linearly independent.

But this requires substantially more machinery.

IDEA:

Consider $\mathbb{R}^{n+1} \cong \mathbb{R} \oplus \mathbb{R}^n$ with product $(a, u)(b, v) = (ab - u \cdot v, av + bu + u \times v)$. Then (1), (2), (3) imply this gives a normed division algebra!!!! So by the above theorem, we must have either $\mathbb{H}$ or $\mathbb{O}$ (since $n \geq 3$, the non ≥ 3 case also works but implies $\times$ is trivial).

SO THE ONLY CROSS PRODUCTS are $\mathbb{R}^3, \mathbb{R}^7$!!! (and trivially 0 in the lower-dimensional Cayley-Dickson). ■